STAT/MA 41600
In-Class Problem Set #25: October 16, 2026
Solutions by Mark Daniel Ward

Problem Set 25 Answers

1a. Because the pdf is constant, $P(X < 4)$ equals the area of the region with $X < 4$ (area $(2.5)(4.0) + (1/2)(2.5)(4.0) = 15$) over the entire area (area $(5)(8)(1/2) = 20$), so $P(X < 4) = 15/20 = 3/4$.

Alternatively, we can integrate $P(X < 4) = \int_0^4 \int_0^{5-5x/8} 1/20 \, dy \, dx = 3/4$.

1b. Because the pdf is constant, $P(X > 2)$ equals the area of the region with $X > 2$ (area $(1/2)(6)(15/4) = 45/4$) over the entire area (area $(5)(8)(1/2) = 20$), so $P(X > 2) = (45/4)/20 = 9/16$.

Alternatively, we can integrate $P(X > 2) = \int_2^8 \int_0^{5-5x/8} 1/20 \, dy \, dx = 9/16$.

2a. We have $P(1 \leq \min(X, Y)) = \int_1^\infty \int_1^\infty 14e^{-2x-7y} \, dy \, dx = e^{-2}e^{-7} = e^{-9}$.

2b. We have $P(3 \leq \min(X, Y)) = \int_3^\infty \int_3^\infty 14e^{-2x-7y} \, dy \, dx = e^{-6}e^{-21} = e^{-27}$.

2c. We have $P(10 \leq \min(X, Y)) = \int_{10}^\infty \int_{10}^\infty 14e^{-2x-7y} \, dy \, dx = e^{-20}e^{-70} = e^{-90}$.

2d. In general, for $a > 0$, we have $P(a \leq \min(X, Y)) = \int_a^\infty \int_a^\infty 14e^{-2x-7y} \, dy \, dx = e^{-2a}e^{-7a} = e^{-9a}$.

3a. We have $P(X + Y \geq 2) = \int_0^2 \int_{2-x}^\infty 14e^{-2x-7y} \, dy \, dx + \int_2^\infty \int_0^\infty 14e^{-2x-7y} \, dy \, dx = \frac{2}{5}(e^{-4} - e^{-14}) + e^{-4} = \frac{7}{5}e^{-4} - \frac{2}{5}e^{-14} = 0.02564$.

Alternatively, we have $P(X+Y \geq 2) = 1 - P(X+Y \leq 2) = 1 - \int_0^2 \int_0^{2-x} 14e^{-2x-7y} dy \, dx = 1 - (1 - \frac{7}{5}e^{-4} + \frac{2}{5}e^{-14}) = \frac{7}{5}e^{-4} - \frac{2}{5}e^{-14} = 0.02564$.

3b. We have $P(X > 4Y) = \int_0^\infty \int_{4y}^\infty 14e^{-2x-7y} \, dx \, dy = 7/15$.

4a. We need $1 = \int_0^5 \int_0^2 k(5 - x)(2 - y) \, dy \, dx = (k)(25)$, and thus $k = 1/25$.

4b. The pdf of X is $f_X(x) = \int_0^2 \frac{1}{25}(5 - x)(2 - y) \, dy = \frac{2}{25}(5 - x)$ for $0 < x < 5$, and $f_X(x) = 0$ otherwise.

4c. The pdf of Y is $f_Y(y) = \int_0^5 \frac{1}{25}(5 - x)(2 - y) \, dx = \frac{1}{2}(2 - y)$ for $0 < y < 2$, and $f_Y(y) = 0$ otherwise.

5a. We need $1 = \int_0^\infty \int_x^\infty ke^{-2x-7y} \, dy \, dx = k/63$, and thus $k = 63$.

5b. The pdf of X is $f_X(x) = \int_x^\infty 63e^{-2x-7y} \, dy = 9e^{-9x}$ for $x > 0$, and $f_X(x) = 0$ otherwise.